

Luohua Yang

Data Scientist / Machine Learning Engineer — 2026 Graduate

Selly Oak, Birmingham, UK • +44 7906 84527 • 1771602053az@gmail.com • [linkedin.com/in/luohua-yang](https://www.linkedin.com/in/luohua-yang) • github.com/VerticalFall

SUMMARY

MSc Statistical Data Science graduate (University of Birmingham) who enjoys building and evaluating machine learning systems end-to-end — from experimental design and modelling, to LLM-assisted pipelines and evaluation. Proficient in Python (scikit-learn, PyTorch) and SQL, experienced in classical ML, deep learning, and statistical learning theory, and familiar with integrating and evaluating multiple LLM providers. Comfortable working in both English and Chinese, with a background in statistics and finance that keeps modelling grounded in real business problems.

SELECTED PROJECTS

stock-monitor — Multi-market Equity Analysis & Daily Report System

2025 – 2026

- Integrated a configurable, multi-provider LLM analysis layer applying Buffett-value and Fisher-growth frameworks with DCF valuation to score and surface candidate stocks across 4 markets.
- Cross-validated DCF growth assumptions against PEG-implied growth, consensus estimates, and historical growth (capped at 50%) to prevent valuation distortion from one-off earnings spikes.
- Instrumented AI-call cost tracking (~\$0.003 per report with the default model) and automated data-integrity checks; owned architecture design and code review while pair-programming with Claude Code.
- Delivered via a Flask + ECharts dashboard with GitHub Actions scheduling and real-time Feishu alerts, cutting manual research time ~90% across 20+ tracked stocks; source at github.com/VerticalFall/stock-monitor.

obsidian-vault — Automated Global-News Knowledge Base & Daily Briefing

2025 – 2026

- Designed a DeepSeek-powered three-layer funnel topic router — functionally a production-style text classification/routing pipeline — distilling 100+ daily articles into a 5-minute bilingual (CN/EN) briefing.
- Engineered a GitHub Actions pipeline aggregating four heterogeneous sources into a structured, LLM-ready corpus; published 160+ daily briefings since July 2026 at zero infrastructure cost.
- Built independent rendering scripts and automated health checks for data integrity and pipeline status.

PROFESSIONAL EXPERIENCE

Data Operations Intern — Tungee (AI sales-intelligence SaaS, ~1,000 staff)

Jun 2024 – Sep 2024

- Designed and implemented a scheduled data pipeline collecting daily KPIs (followers, views, comments) for 50+ KOL accounts on Douyin and Bilibili, replacing manual daily logging.
- Developed interactive Tableau dashboards visualising performance trends and audience engagement patterns, used in weekly content reviews.
- Fed data-backed insights into content strategy, helping the operations team identify top-performing content formats ~2x faster.

Research Project Member — Guangdong University of Foreign Studies

2021 – 2025

- Owned data collection for Southeast Asia sources (Thailand, Indonesia, Singapore): tourism/culture portals and government news sites.
- Built a scheduled weekly incremental crawler (requests, MySQL) feeding full news content into a preference/labelling pipeline; upgraded the system for reliability and coverage.

EDUCATION

University of Birmingham — MSc, Statistical Data Science (Distinction expected)

Sep 2025 – Dec 2026

- In progress; expected Dec 2026 (degree conferral date).
- Thesis "Sentiment Analysis of Financial News": compared generative (Naive Bayes) vs. discriminative (linear SVM) classifiers on the Financial PhraseBank dataset (4,846 sentences, 3-class, labelled by 16 finance experts) to empirically test the Ng–Jordan learning-curve theory — generative model ahead under small samples (crossover $n \approx 256$ –512), SVM best on the full dataset (accuracy 0.757 vs 0.698; macro-F1 0.691 vs 0.515).
- Ran robustness experiments on annotation-noise sensitivity and class imbalance, using class-prior/weight adjustment to raise minority (negative-class) recall from 0.151 to 0.456.
- Coursework in machine learning, statistics, deep learning, and econometrics.

Guangdong University of Foreign Studies — BSc, Statistics

2021 – 2025

- Key topics: probability theory, mathematical statistics, econometrics.

CORE COMPETENCIES

ML / AI: Classification & sentiment analysis (SVM, Naive Bayes, deep learning), statistical learning theory, learning-curve analysis; integrating and evaluating multiple mainstream LLM providers

Programming: Python (pandas, NumPy, scikit-learn, PyTorch/TensorFlow), SQL, JavaScript

Statistics: Hypothesis testing, statistical inference, class-imbalance handling

Data Ops / Visualisation: Web scraping (requests), scheduled pipelines (GitHub Actions, cron), MySQL, REST API integration, Tableau, ECharts, BI dashboards

Tools: Git, Linux, Docker, Flask, Feishu/Lark API, Claude Code | **Languages:** Chinese (native), English (professional working proficiency)